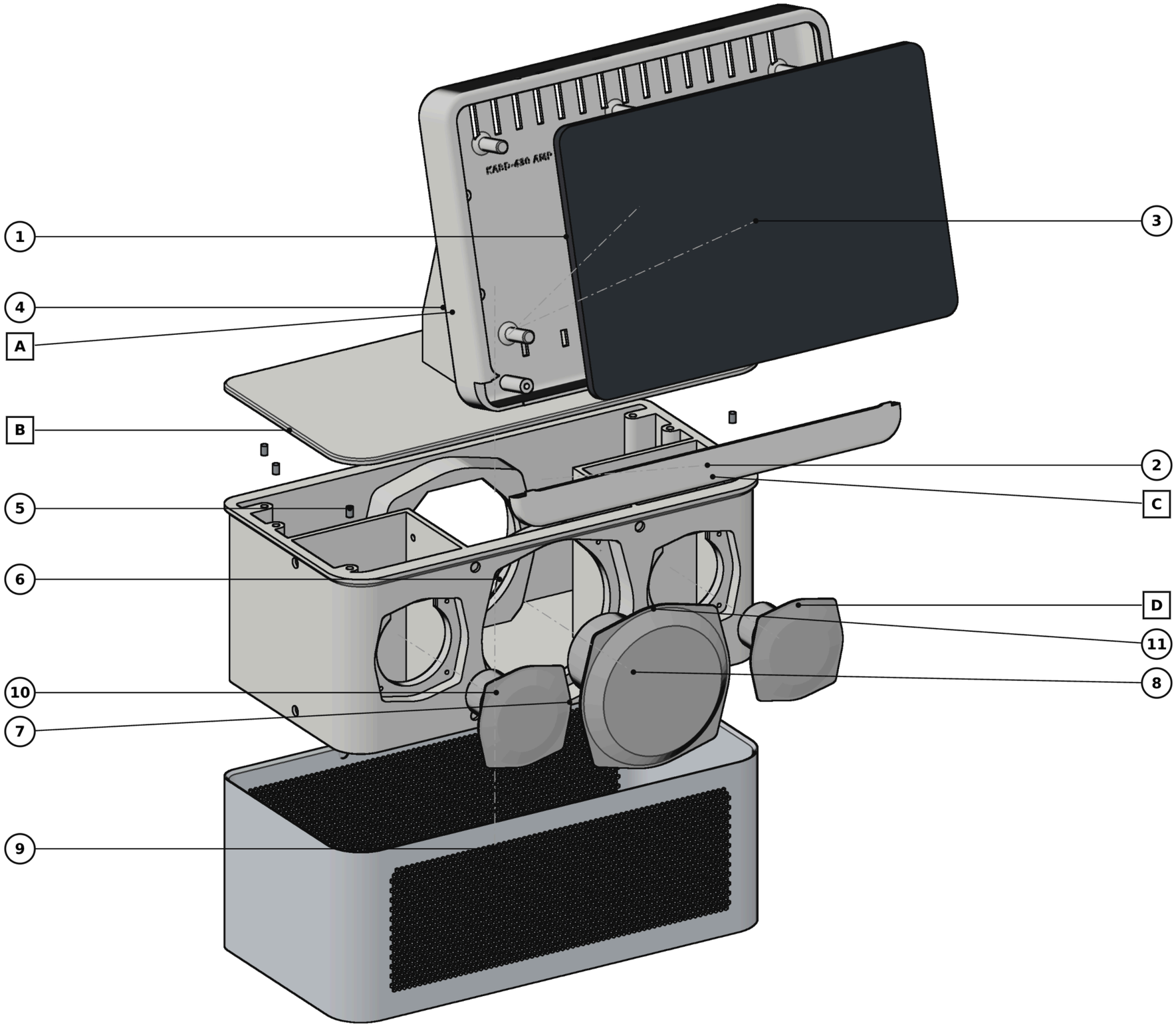


GENERAL ASSEMBLY

Exploded view, front

ITEM	DESCRIPTION	QTY	BOM
1	Bezel shell	1	P4
2	Chin bar	1	P5
3	Tablet, Galaxy Tab A9+	1	E1
4	Lid and spine	1	P2
5	Dowel pin, 4 x 6 mm	6	H5
6	Base tub	1	P1
7	Full-range driver, ND65-4	2	S1
8	Woofer, ND105-4	1	S2
9	Grille shell (or grille ring + linen)	1	P3a
10	Driver screw, M3 x 6 thread-forming	16	S4
11	Gasket tape, 1 mm (under each flange)	1	S5

NOTES

- **A** Bezel bolts to the spine from the rear (sheet 2).
- **B** Epoxy the lid to the tub: outer rim and both full-range chamber walls. Airtight.
- **C** Tablet slides up into the shell from below, then the chin bar closes the slot.
- **D** ND65: tabs up, through the opening above the cutout. Solder after mounting, then seal the opening from inside.

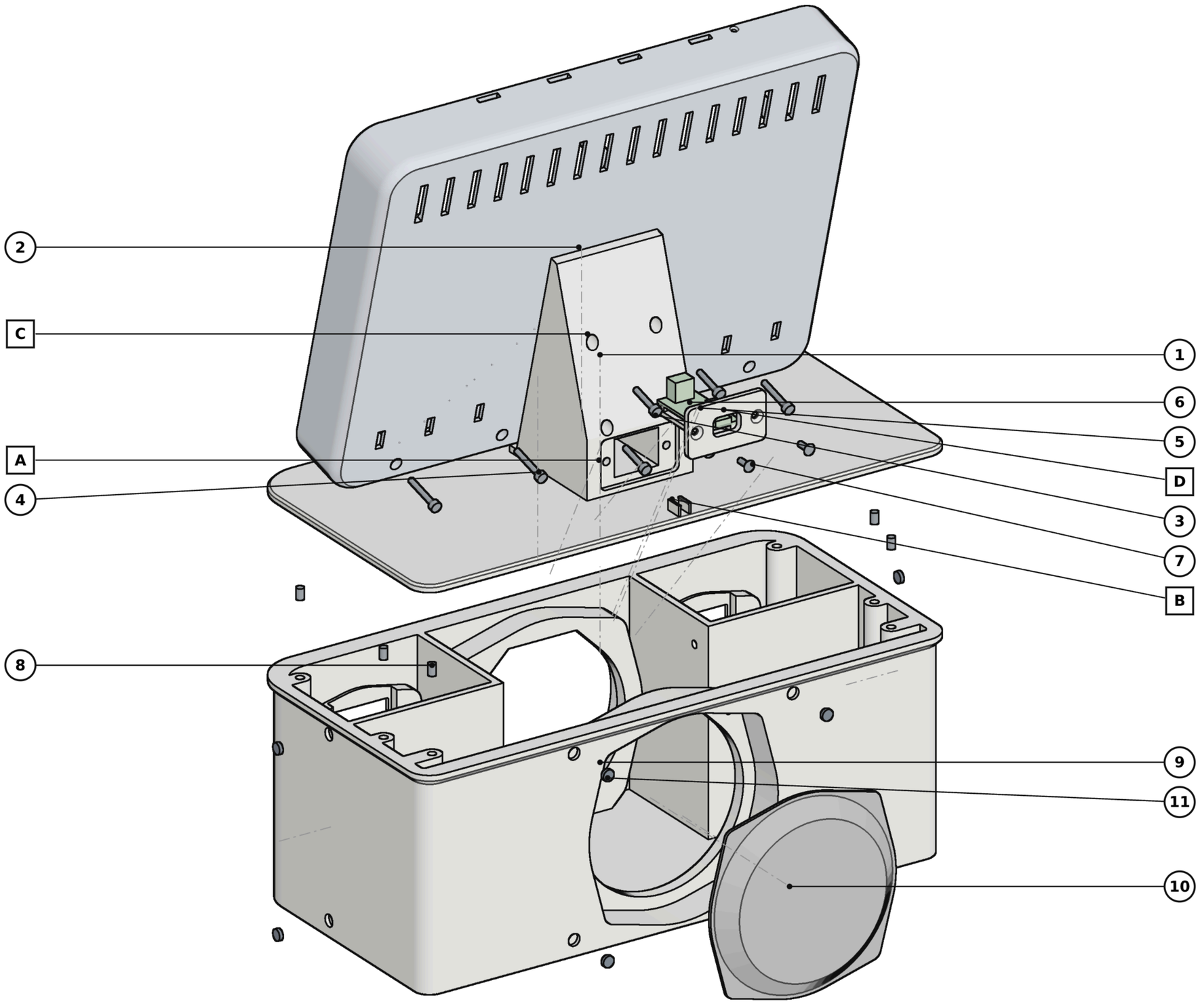
SMARTHUB · COUNTER DISPLAY

DRAWING
GENERAL ASSEMBLY

DWG NO. SH-ASM-01	SHEET 1 OF 4
REV A	DATE 2026-11-05
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BASE AND SPINE

Exploded view, rear



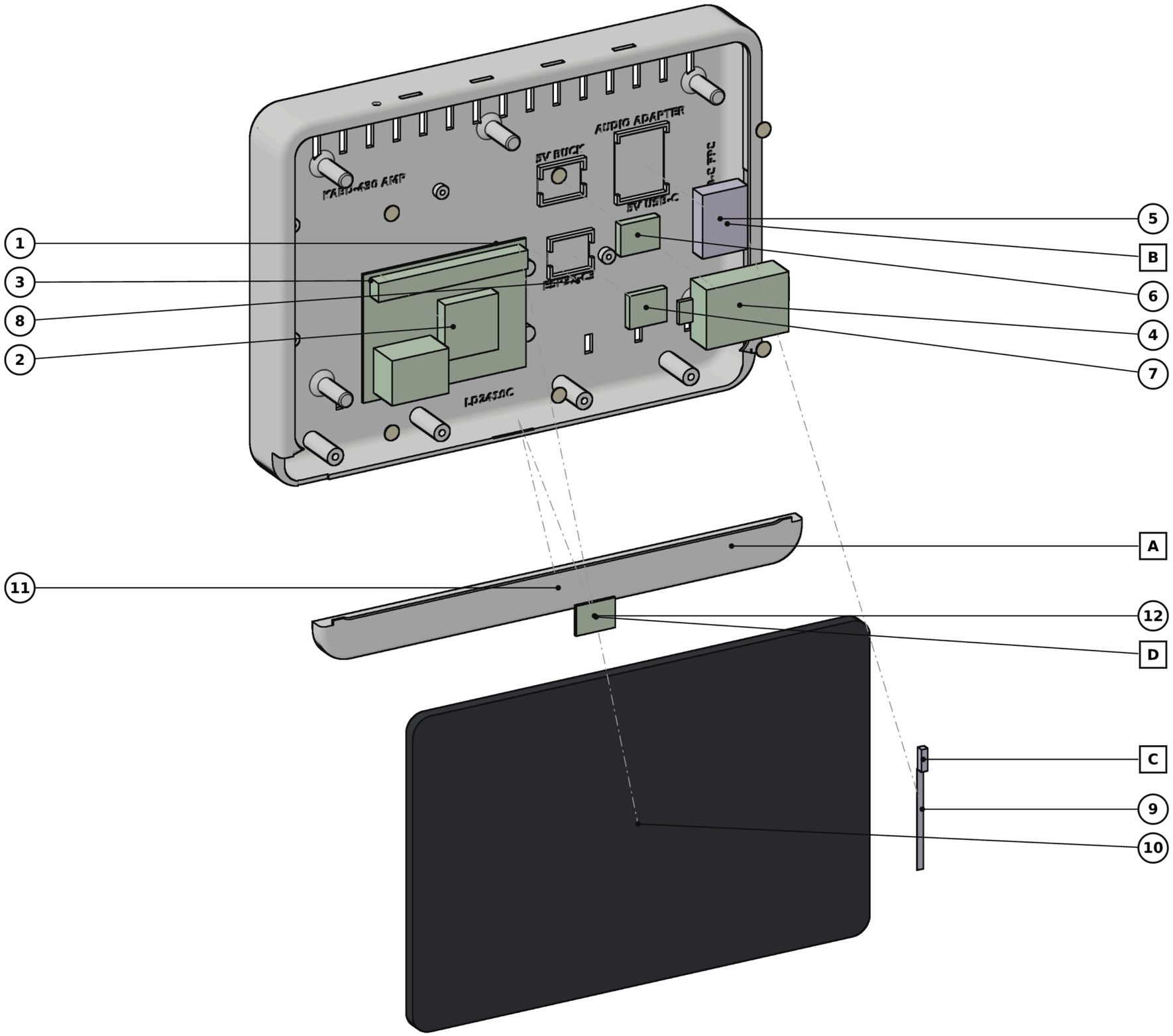
ITEM	DESCRIPTION	QTY	BOM
1	Lid and spine	1	P2
2	Bezel assembly (sheet 3)	1	
3	Bezel bolt, M3 x 16 socket head	4	H3
4	Chin bar screw, M3 x 20 socket head	4	H2
5	Power plate	1	P6
6	PD trigger, HUNSOOL (DIP = 20 V)	1	E9
7	Plate screw, M3 x 8 countersunk	2	H3a
8	Dowel pin, 4 x 6 mm	6	H5
9	Base tub	1	P1
10	Passive radiator, ND105-PR	1	S3
11	Magnet, 6 x 2 mm (tub pockets)	12	H6

NOTES

- **A** Ø8 speaker wire hole in the lid, under the spine. Fill it with hot glue from inside before the lid goes on.
- **B** Cable clip: press the charger cable in here.
- **C** Speaker and 20 V leads run up the spine channel into the tray.
- **D** Set the DIP switch to 20 V only before fitting the plate.

SMARTHUB · COUNTER DISPLAY

DRAWING **BASE AND SPINE**	
DWG NO. SH-ASM-02	SHEET 2 OF 4
REV A	DATE 2026-11-05
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BEZEL ASSEMBLY

Exploded view, front (bezel frame, upright)

ITEM	DESCRIPTION	QTY	BOM
1	Bezel shell	1	P4
2	Amplifier, Dayton KABD-430	1	E2
3	Amp screw, M3 x 6 thread-forming	4	H4
4	5 V USB-C converter (charges the tablet)	1	E6
5	UGREEN 2-in-1 USB-C audio + PD adapter	1	A2
6	5 V buck, MP1584EN	1	E5
7	Seeed XIAO ESP32-C3	1	E3
8	Felt pad, 8 mm, 0.5 mm thick	6	H9
9	Flat FPC USB-C 90° adapter	1	A1
10	Tablet, Galaxy Tab A9+	1	E1
11	Chin bar	1	P5
12	Radar, HLK-LD2410C	1	E4

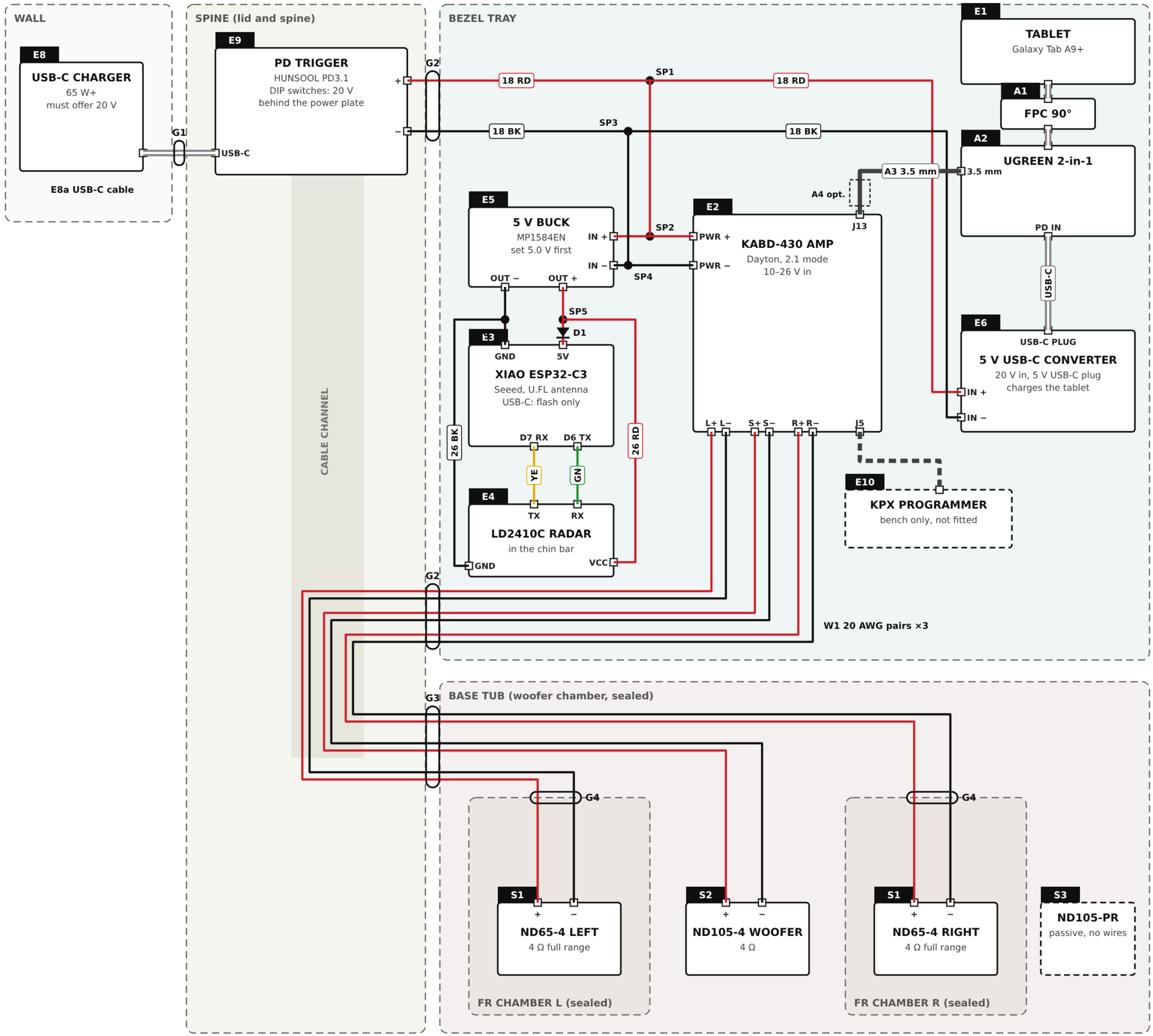
NOTES

- **A** Heat-set inserts: 4 in the chin bar's back face, 4 in the spine-bolt bosses (from the back).
- **B** Small boards sit in their labeled cradles on VHB tape.
- **C** Plug the FPC adapter into the tablet first. It rides in the channel on the right as the tablet slides up into the shell.
- **D** Radar looks out through 1.6 mm of the chin. No metal-filled filament.

SMARTHUB · COUNTER DISPLAY

DRAWING
BEZEL ASSEMBLY

DWG NO.	SHEET
SH-ASM-03	3 OF 4
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SCALE	SOURCE
NTS	enclosure.py 61e43d7



WIRING DIAGRAM

Power, audio and sensor circuits

ID	COMPONENT	WHERE
E8	USB-C PD charger, 65 W+	wall
E9	PD trigger, HUNSOOL (DIP 20 V)	spine
E2	KABD-430 amplifier	tray
E6	5 V USB-C converter	tray
E5	5 V buck, MP1584EN	tray
E3	Seeed XIAO ESP32-C3	tray
E4	LD2410C radar	chin
A1 / A2	FPC 90° adapter / UGREEN 2-in-1	tray
A3 / A4	3.5 mm cable / ground loop isolator (opt.)	tray
S1–S3	ND65-4 ×2, ND105-4, ND105-PR	tub
E10	KPX programmer (to J5, bench only)	—

WIRE	CODE	USE
	18 RD	+20 V (W2)
	18 BK	Ground (W2)
	20 pair	Speaker, + / − (W1)
	26 RD / BK	5 V / ground (W3)
	26 YE / GN	Radar TX → D7 (IO20), RX ← D6 (IO21)
	USB-C	Cables and adapters
	3.5 mm	Line-level audio

PASS-THROUGHS

- **G1** Power plate opening on the spine's rear.
- **G2** Ø13 cable hole in the tray's back wall, at the top of the spine channel.
- **G3** Ø8 lid hole into the woofer chamber. **Fill with hot glue.**
- **G4** Full-range chamber wall holes ×2. **Fill with hot glue.**

NOTES

1. Check for 20 V at E9's terminal before connecting anything.
2. If the buck is adjustable, set it to 5.0 V before connecting E3 and E4.
3. Output channels follow the DSP program loaded through J5. Check the board's silkscreen. J13 is the AUX input.
4. Speaker + goes to each driver's red-marked terminal.
5. Fit A4 only if you hear hum while the tablet charges.
6. D1 is a Schottky diode (1N5819 or SS14) into the XIAO's 5V pin, so USB can't back-feed the buck. Plug in the XIAO's antenna.

SMARTHUB · COUNTER DISPLAY

DRAWING **WIRING DIAGRAM**	
DWG NO. SH-WIR-01	SHEET 4 OF 4
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